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SHEET CONVEYING DEVICE, IMAGE PROCESSING APPARATUS, AND DRIVE CONTROL METHOD CAPABLE OF SUPPRESSING TEMPERATURE RISE OF MOTOR WITH SIMPLE CONFIGURATION

Abstract

A sheet conveying device includes a motor, a first determination processing portion, and a first restriction processing portion. The motor is used to convey a sheet. The first determination processing portion determines, each time a predetermined determination timing arrives during drive of the motor, whether or not a total drive time of the motor in a predetermined determination period up to the arrival of the determination timing exceeds a predetermined first time. The first restriction processing portion restricts the drive of the motor when the first determination processing portion determines that the total drive time exceeds the first time.

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Background/Summary

INCORPORATION BY REFERENCE

[0001] This application is based upon and claims the benefit of priority from the corresponding Japanese Patent Application No. 2024-033454 filed on Mar. 6, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND

[0002] The present disclosure relates to a sheet conveying device, an image processing apparatus, and a drive control method.

[0003] There is known an image processing apparatus including an image reading portion for reading an image of a document sheet and a document sheet conveying portion for conveying a document sheet whose image is read by the image reading portion. In addition, an image processing apparatus capable of suppressing a temperature rise of a motor for driving the document sheet conveying portion is known as a related art.

SUMMARY

[0004] A sheet conveying device according to one aspect of the present disclosure includes a motor, a first determination processing portion, a first restriction processing portion, a second determination processing portion, and a second restriction processing portion. The motor is used to convey a sheet. The first determination processing portion determines, each time a predetermined determination timing arrives during drive of the motor, whether or not a total drive time of the motor in a predetermined determination period up to the arrival of the determination timing exceeds a predetermined first time. The first restriction processing portion restricts the drive of the motor when the first determination processing portion determines that the total drive time exceeds the first time. The second determination processing portion determines whether or not an elapsed time from start of the drive of the motor immediately before a reference time at which the drive of the motor is started to the reference time exceeds a length of the determination period. The second restriction processing portion restricts the drive of the motor when the second determination processing portion determines that the elapsed time exceeds the length of the determination period and a time of the drive of the motor from the reference time exceeds a second time that is shorter than the first time.

[0005] An image processing apparatus according to another aspect of the present disclosure includes the sheet conveying device and an image reading portion. The image reading portion reads an image of the sheet conveyed by the sheet conveying device.

[0006] A drive control method according to another aspect of the present disclosure is performed in a sheet conveying device including a motor used to convey a sheet, and includes a first determination step, a first restriction step, a second determination step, and a second restriction step. In the first determination step, each time a predetermined determination timing arrives during drive of the motor, it is determined whether or not a total drive time of the motor in a predetermined determination period up to the arrival of the determination timing exceeds a predetermined first time. In the first restriction step, the drive of the motor is restricted when it is determined in the first determination step that the total drive time exceeds the first time. In the second determination step, it is determined whether or not an elapsed time from start of the drive of the motor immediately before a reference time at which the drive of the motor is started to the reference time exceeds a length of the determination period. In the second restriction step, the drive of the motor is restricted when it is determined in the second determination step that the elapsed time exceeds the length of the determination period and a time of the drive of the motor from the

reference time exceeds a second time that is shorter than the first time.

[0007] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description with reference where appropriate to the accompanying drawings. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Furthermore, the claimed subject matter is not limited to implementations that solve any or all disadvantages noted in any part of this disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a block diagram showing a system configuration of an image forming apparatus according to an embodiment of the present disclosure.

[0009] FIG. 2 is a diagram showing configurations of an ADF and an image reading portion of the image forming apparatus according to the embodiment of the present disclosure.

[0010] FIG. 3 is a timing chart showing an example of a change with time in the drive state of the motor of the image forming apparatus according to the embodiment of the present disclosure.

[0011] FIG. 4 is a timing chart showing an example of a change with time in the drive state of the motor of the image forming apparatus according to the embodiment of the present disclosure.

[0012] FIG. 5 is a timing chart showing an example of a change with time in the drive state of the motor of the image forming apparatus according to the embodiment of the present disclosure.

[0013] FIG. 6 is a flowchart showing an example of drive control processing executed in the image forming apparatus according to the embodiment of the present disclosure.

DETAILED DESCRIPTION

[0014] Hereinafter, an embodiment of the present disclosure will be described with reference to the accompanying drawings. It is noted that the following embodiment is an example of embodying the present disclosure and does not limit the technical scope of the present disclosure.

Configuration of Image Forming Apparatus 100

[0015] First, a configuration of an image forming apparatus 100 according to the embodiment of the present disclosure will be described with reference to FIG. 1.

[0016] The image forming apparatus 100 is a multifunction peripheral having a plurality of functions such as a facsimile function and a copy function in addition to a scan function for reading an image from a document sheet and a print function for forming an image based on image data. The image forming apparatus 100 is an example of the image processing apparatus of the present disclosure. It is noted that the image processing apparatus of the present disclosure may be a scanner, a printer, a facsimile machine, a copier, or the like.

[0017] As shown in FIG. 1, the image forming apparatus 100 includes an auto document feeder (ADF) 1, an image reading portion 2, an image forming portion 3, a sheet feed portion 4, an operation display portion 5, a storage portion 6, and a control portion 7.

[0018] The ADF 1 conveys a document sheet to be read (an example of the sheet of the present disclosure). The ADF 1 executes conveying processing by the driving force supplied from a motor 27A in accordance with a control instruction from the control portion 7.

[0019] The image reading portion 2 reads an image of a document sheet conveyed by the ADF 1. In addition, the image reading portion 2 reads an image of a document sheet placed on a document sheet table 41 (see FIG. 2).

[0020] The image forming portion 3 forms an image on a sheet supplied from the sheet feed portion 4, using electrophotographic method. For example, the image forming portion 3 includes a photoconductor drum, a charging roller, a laser scanning unit, a developing device, a transfer roller, a cleaning device, a fixing device, and a sheet discharge tray.

[0021] The sheet feed portion **4** supplies a sheet to the image forming apparatus **100**. For example, the sheet feed portion **4** includes a sheet feed cassette, a manual feed tray, a sheet conveying path, and a plurality of conveying rollers.

[0022] The operation display portion **5** is a user interface of the image forming apparatus **100**. For example, the operation display portion **5** includes a display portion such as a liquid crystal display that displays various types of information in response to control instructions from the control portion **7**, and an operation portion such as operation keys or a touch panel that inputs various types of information to the control portion **7** in response to user operations.

[0023] The storage portion **6** is a nonvolatile storage device. For example, the storage portion **6** is a storage device such as a flash memory or other nonvolatile memory, a solid state drive (SSD), or a hard disk drive (HDD).

[0024] The control portion **7** performs overall control of the image forming apparatus **100**. As shown in FIG. **1**, the control portion **7** includes a CPU **11**, a ROM **12**, and a RAM **13**. The CPU **11** is a processor that executes various types of arithmetic processing. The ROM **12** is a nonvolatile storage device in which information such as control programs for causing the CPU **11** to execute various types of processing are stored in advance. The RAM **13** is a volatile or nonvolatile storage device used as a temporary storage memory (work area) for various types of processing executed by the CPU **11**. In the control portion **7**, various control programs stored in the ROM **12** in advance are executed by the CPU **11**. The control portion **7** thereby performs overall control of the image forming apparatus **100**.

Configurations of ADF **1** and Image Reading Portion **2**

[0025] Next, configurations of the ADF **1** and the image reading portion **2** will be described with reference to FIG. **1** and FIG. **2**. Here, FIG. **2** is a cross-sectional view showing the configurations of the ADF **1** and the image reading portion **2**.

[0026] As shown in FIG. **2**, the ADF **1** includes a document sheet mounting portion **21**, a housing **22**, a pickup roller **23**, a sheet feed belt **24**, a separation roller **25**, a registration roller **26**, a motor **27A**, a first conveying roller **30**, a second conveying roller **31**, a discharge roller **32**, and a discharge portion **33**.

[0027] A document sheet to be conveyed is placed on the document sheet mounting portion **21**. In the ADF **1**, the document sheet placed on the document sheet mounting portion **21** is conveyed in a conveying direction **D1** shown in FIG. **2**.

[0028] The document sheet mounting portion **21** is provided with a detection portion **21A** shown in FIG. **1**. The detection portion **21A** detects the presence or absence of a document sheet on the document sheet mounting portion **21**. For example, the detection portion **21A** is a reflective optical sensor provided on a document sheet mounting surface of the document sheet mounting portion **21**.

[0029] The document sheet mounting portion **21** is provided with a lift plate (not shown). The lift plate lifts a document sheet stack placed on the document sheet mounting portion **21** to a position where the document sheet stack comes into contact with the pickup roller **23**.

[0030] The housing **22** houses a roller or the like used to convey a document sheet. As shown in FIG. **2**, a conveying path **22A** for guiding a document sheet is formed inside the housing **22**.

[0031] In the ADF **1**, the document sheet placed on the document sheet mounting portion **21** is conveyed to the discharge portion **33** through the conveying path **22A**.

[0032] As shown in FIG. **2**, an opening **22B** is provided at the bottom of the housing **22**. The opening **22B** exposes a part of the conveying path **22A** to the outside. In the opening **22B**, a first imaging portion **42** (see FIG. **1** and FIG. **2**) of the image reading portion **2** reads the image of the document sheet conveyed by the ADF **1**.

[0033] The pickup roller **23** is provided above the document sheet mounting portion **21**. The pickup roller **23** comes into contact with the surface (upper surface) of the uppermost document sheet in the document sheet stack lifted by the lift plate, and conveys the document sheet in the conveying direction **D1**.

[0034] The sheet feed belt **24** is provided on the conveying path **22A**. The sheet feed belt **24** comes into contact with the surface of the document sheet conveyed in the conveying direction **D1** by the pickup roller **23**, and conveys the document sheet downstream in the conveying direction **D1**.

[0035] The separation roller **25** is provided below the sheet feed belt **24** in contact with the sheet feed belt **24**. The separation roller **25** separates a document sheet that is in contact with the sheet feed belt **24** from the document sheet below it among a plurality of document sheets conveyed to a contact position with the sheet feed belt **24**.

[0036] The registration roller **26** is provided downstream of the sheet feed belt **24** in the conveying direction **D1** on the conveying path **22A**. The registration roller **26** comes into contact with the document sheet conveyed by the sheet feed belt **24**, and conveys the document sheet downstream in the conveying direction **D1**.

[0037] The first conveying roller **30** is provided downstream of the registration roller **26** in the conveying direction **D1** on the conveying path **22A**. The first conveying roller **30** comes into contact with a document sheet and conveys the document sheet downstream in the conveying direction **D1**.

[0038] An opening **22B** is formed downstream of the first conveying roller **30** in the conveying direction **D1** on the conveying path **22A**. At the opening **22B**, the image of the document sheet passing through the opening **22B** is read by the first imaging portion **42** (see FIG. 1 and FIG. 2) of the image reading portion **2**.

[0039] The second conveying roller **31** is provided downstream of the opening **22B** in the conveying direction **D1** on the conveying path **22A**. The second conveying roller **31** comes into contact with a document sheet conveyed by the first conveying roller **30**, and conveys the document sheet downstream in the conveying direction **D1**.

[0040] The discharge roller **32** is provided downstream of the second conveying roller **31** in the conveying direction **D1** on the conveying path **22A**. The discharge roller **32** comes into contact with a document sheet conveyed by the second conveying roller **31** and discharges the document sheet to the discharge portion **33**. Document sheets discharged by the discharge roller **32** is stacked on the discharge portion **33**.

[0041] As shown in FIG. 1 and FIG. 2, the image reading portion **2** includes a document sheet table **41**, a first imaging portion **42**, and a second imaging portion **43**.

[0042] A document sheet to be read is placed on the document sheet table **41**. The document sheet table **41** is provided at the top of the housing of the image forming apparatus **100**. The ADF **1** is provided so as to be openable and closable with respect to the document sheet table **41**, and also serves as a document sheet cover for covering one surface of the document sheet placed on a first contact glass **411** of the document sheet table **41**.

[0043] As shown in FIG. 2, the document sheet table **41** includes a first contact glass **411**, a second contact glass **412**, and a guide member **413**. A document sheet whose image is to be read by the first imaging portion **42** is placed on the first contact glass **411**. When the ADF **1** is closed with respect to the document sheet table **41**, the second contact glass **412** and the guide member **413** face the opening **22B** of the housing **22** to form a part of the conveying path **22A**. The second contact glass **412** transmits light emitted from the first imaging portion **42** toward the opening **22B** and light reflected by the document sheet. The guide member **413** guides the document sheet to the second conveying roller **31** downstream of the second contact glass **412** in the conveying direction **D1**.

[0044] The first imaging portion **42** is provided below the first contact glass **411** and the second contact glass **412**. The first imaging portion **42** is long in the depth direction of the paper surface in FIG. 2, and is provided so as to be movable in the left-right direction of the paper surface in FIG. 2. The first imaging portion **42** reads an image of the front surface of the document sheet conveyed by the ADF **1** in a state where it is disposed below the second contact glass **412**. Specifically, the first imaging portion **42** includes a contact image sensor (CIS) and a housing that houses the CIS. The

first imaging portion **42** outputs an analog signal corresponding to the image read from the front surface of the document sheet. The analog signal output from the first imaging portion **42** is converted into a digital signal (image data) by an analog front-end circuit (not shown), and is input to the control portion **7**.

[0045] The second imaging portion **43** is provided downstream of the registration roller **26** in the conveying direction **D1** and upstream of the first conveying roller **30** in the conveying direction **D1** on the conveying path **22A**. The second imaging portion **43** reads an image of the back surface of the document sheet conveyed by the registration roller **26**. Specifically, the second imaging portion **43** is a CIS. The second imaging portion **43** outputs an analog signal corresponding to the image read from the back surface of the document sheet. The analog signal output from the second imaging portion **43** is converted into a digital signal (image data) by the analog front-end circuit, and is input to the control portion **7**.

[0046] The ADF **1** operates by receiving a driving force supplied from a motor **27A** shown in FIG. **1**. That is, the motor **27A** is used to convey a sheet. The motor **27A** outputs the driving force to the ADF **1** in accordance with a control instruction from the control portion **7**. In addition, the motor **27A** is set to an ON state and an OFF state in accordance with a control instruction from the control portion **7**.

[0047] By the way, when the motor **27A** is continuously driven (operated) for a long time, the temperature of the motor **27A** may rise, leading to a failure. Here, an image processing apparatus capable of suppressing the temperature rise of the motor **27A** is known as a related art.

[0048] However, in the image processing apparatus according to the related art, the drive of the motor **27A** is controlled based on the prediction result of the temperature of the motor **27A**. Therefore, a configuration for predicting the temperature of the motor **27A** is required, which makes the configuration for suppressing the temperature rise of the motor **27A** more complicated.

[0049] On the other hand, in the image forming apparatus **100** according to the embodiment of the present disclosure, as will be described below, the temperature rise of the motor **27A** can be suppressed with a simple configuration.

Configuration of Control Portion **7**

[0050] Next, the configuration of the control portion **7** will be described in more detail below with reference to FIG. **1**. As shown in FIG. **1**, the control portion **7** includes a recording processing portion **51**, a first determination processing portion **52**, a first restriction processing portion **53**, a second determination processing portion **54**, and a second restriction processing portion **55**.

[0051] Specifically, a drive control program for causing the CPU **11** to execute drive control processing (see FIG. **6**) to be described later is stored in the ROM **12** in advance. By executing the drive control programs stored in the ROM **12**, the control portion **7** functions as the processing portions described above. Here, the device including the ADF **1** and the control portion **7** is an example of the sheet conveying device of the present disclosure.

[0052] It is noted that the drive control programs may be recorded on a computer-readable recording medium such as a CD, a DVD, or a flash memory, and may be read from the recording medium and stored in a storage device such as the storage portion **6**. In addition, some or all of the processing portions included in the control portion **7** may be constituted by electronic circuits. In addition, the drive control programs may be programs for causing a plurality of processors to function as the processing portions included in the control portion **7**.

[0053] The recording processing portion **51** records a change in the drive state of the motor **27A**.

[0054] For example, when the drive of the motor **27A** is started (changed from the OFF state to the ON state), the recording processing portion **51** records the drive start time of the motor **27A**. Specifically, the recording processing portion **51** stores information indicating the drive start time of the motor **27A** in a predetermined storage area of the storage portion **6**.

[0055] When the drive of the motor **27A** is ended (changed from the ON state to the OFF state), the recording processing portion **51** records the drive end time of the motor **27A**. Specifically, the

recording processing portion **51** stores information indicating the drive end time of the motor **27A** in the storage area of the storage portion **6**.

[0056] It is noted that the recording processing portion **51** may determine the current drive state of the motor **27A** in a predetermined recording cycle, and store information indicating the determination result in the storage area of the storage portion **6**.

[0057] Each time a predetermined determination timing arrives during the drive of the motor **27A**, the first determination processing portion **52** determines whether or not the total drive time of the motor **27A** in a predetermined determination period **Z10** (see FIG. 3) up to the arrival of the determination timing exceeds a predetermined first time **X1** (see FIG. 3). It is noted that the determination period **Z10** is indicated by a dash-dot-dot-dash line in FIG. 3.

[0058] For example, the determination timing is a timing that comes every 10 seconds from the start of the drive of the motor **27A**. It is noted that the determination timing may be any timing that comes periodically or irregularly during the drive of the motor **27A**.

[0059] In addition, the determination period **Z10** has a length **X10** (see FIG. 3) of 28 minutes.

[0060] In addition, the first time **X1** is 18.2 minutes.

[0061] For example, when the determination timing arrives, the first determination processing portion **52** acquires the total drive time of the motor **27A** in the determination period **Z10** based on the recording result of the recording processing portion **51**. Then, the first determination processing portion **52** determines whether or not the acquired total drive time of the motor **27A** exceeds the first time **X1**.

[0062] For example, it is assumed that the motor **27A** is driven as shown in FIG. 3 before the determination timing **T12** shown in FIG. 3. In this case, the first determination processing portion **52** acquires the elapsed time from the drive start timing **T11** to the determination timing **T12** as the total drive time of the motor **27A** in the determination period **Z10**. Then, the first determination processing portion **52** determines that the acquired total drive time of the motor **27A** exceeds the first time **X1**.

[0063] When the continuous drive time exceeds 18.2 minutes, the motor **27A** shifts to an abnormal state in which the temperature of the motor **27A** exceeds a predetermined upper limit temperature. When the motor **27A** is driven in the abnormal state, the risk of failure of the motor **27A** increases. In addition, when the motor **27A** is driven in the abnormal state, problems such as the motor **27A** igniting, or members around the motor **27A** being deformed due to heat generated by the motor **27A** occur. Therefore, in the image forming apparatus **100**, the first time **X1** is set to 18.2 minutes.

[0064] The first restriction processing portion **53** restricts the drive of the motor **27A** when the first determination processing portion **52** determines that the total drive time of the motor **27A** in the determination period **Z10** exceeds the first time **X1**.

[0065] For example, the first restriction processing portion **53** switches the document sheet conveying mode of the ADF **1** from a normal conveying mode to a low-speed conveying mode in which the number of conveyed document sheets per minute is smaller than that in the normal conveying mode. In the normal conveying mode, a plurality of document sheets are continuously conveyed without the motor **27A** being stopped. On the other hand, in the low-speed conveying mode, the motor **27A** is stopped for a certain period of time each time a document sheet is conveyed. In other words, the motor **27A** is intermittently driven (see FIG. 3). For example, in the low-speed conveying mode, the motor **27A** is stopped for four seconds each time a document sheet reaches the first conveying roller **30** (see FIG. 2). Thus, the temperature rise of the motor **27A** caused by the continuous drive of the motor **27A** is suppressed.

[0066] In addition, when the conveying mode is shifted to the low-speed conveying mode, the first restriction processing portion **53** determines whether or not the total drive time of the motor **27A** in the determination period **Z10** is equal to or less than a third time (for example, 16.2 minutes), which is shorter than the first time **X1**, every time the determination timing arrives. When it is determined that the total drive time of the motor **27A** in the determination period **Z10** is equal to or

less than the third time, the first restriction processing portion **53** switches the conveying mode from the low-speed conveying mode to the normal conveying mode. In other words, the first restriction processing portion **53** lifts the restriction of the drive of the motor **27A**. When the low-speed conveying mode continues for more than a predetermined time (for example, 10 minutes), the first restriction processing portion **53** switches the conveying mode from the low-speed conveying mode to the normal conveying mode.

[0067] It is noted that, when the first determination processing portion **52** determines that the total drive time of the motor **27A** in the determination period **Z10** exceeds the first time **X1**, the first restriction processing portion **53** may stop the motor **27A** for a predetermined time instead of switching the conveying mode to the low-speed conveying mode.

[0068] It is now assumed that the motor **27A** is driven as shown in FIG. 4 before the drive start timing **T23** shown in FIG. 4. In this case, when the drive time of the motor **27A** from the drive start timing **T23** included in the determination period **Z10** increases, the drive time of the motor **27A** up to the drive end timing **T22** included in the determination period **Z10** decreases by the same amount as the increase. Therefore, the drive of the motor **27A** is not restricted until the continuous drive time of the motor **27A** from the drive start timing **T23** exceeds the first time **X1**.

[0069] However, at the drive start timing **T23**, the motor **27A** may not have cooled to the temperature of the installation environment of the image forming apparatus **100**. That is, the heat generated during the immediately preceding drive period of the motor **27A** (the period from the drive start timing **T21** to the drive end timing **T22**) may remain. In this case, if the drive of the motor **27A** is not restricted until the continuous drive time of the motor **27A** from the drive start timing **T23** exceeds the first time **X1**, the motor **27A** shifts to the abnormal state before the continuous drive time of the motor **27A** from the drive start timing **T23** reaches the first time **X1**, and the risk of failure of the motor **27A** increases.

[0070] On the other hand, in the image forming apparatus **100** according to the embodiment of the present disclosure, as will be described below, the risk of failure of the motor **27A** can be reduced.

[0071] The second determination processing portion **54** determines whether or not the elapsed time from the start of the drive of the motor **27A** immediately before the reference time at which the drive of the motor **27A** is started to the reference time exceeds the length **X10** of the determination period **Z10**. The reference time is when the motor **27A** starts to be driven.

[0072] For example, it is assumed that the motor **27A** is driven as shown in FIG. 4 until the drive start timing **T23** shown in FIG. 4. In this case, the second determination processing portion **54** determines that the elapsed time from the start of the drive of the motor **27A** immediately before the drive start timing **T23**, which is the reference time (the elapsed time from the drive start timing **T21** to the drive start timing **T23**) exceeds the length **X10** of the determination period **Z10**.

[0073] The second restriction processing portion **55** restricts the drive of the motor **27A** when the second determination processing portion **54** determines that the elapsed time from the start of the drive of the motor **27A** immediately before the reference time to the reference time exceeds the length **X10** of the determination period **Z10** and the drive time of the motor **27A** from the reference time exceeds a second time **X2** (see FIG. 5), which is shorter than the first time **X1**.

[0074] For example, the second time **X2** is set based on the difference between the length **X10** of the determination period **Z10** and the stop time of the motor **27A** up to the reference time, and the first time **X1**. For example, the second time **X2** is obtained by subtracting the difference between the length **X10** of the determination period **Z10** and the stop time of the motor **27A** up to the reference time (the elapsed time from the drive end timing **T22** to the drive start timing **T23**) from the first time **X1** (see FIG. 4). By setting the second time **X2** as described above, the second time **X2** can be set in consideration of the elapsed time (natural cooling time of the motor **27A**) from the end of the drive of the motor **27A** immediately before the reference time.

[0075] It is noted that the second time **X2** may be a time obtained by multiplying the first time **X1** by a predetermined coefficient (for example, 0.5).

Drive Control Processing

[0076] The drive control method of the present disclosure will be described below with reference to FIG. 6, along with an example of the procedure of the drive control processing executed by the control portion 7 in the image forming apparatus 100. Here, steps S11, S12, . . . represent the numbers of the processing procedure (steps) executed by the control portion 7. It is noted that, when the conveyance of a document sheet by the ADF 1 (the conveying processing) is started, the drive control processing is executed together with the conveying processing.

[0077] It is noted that the change in the drive state of the motor 27A during the execution of the conveying processing is recorded by the recording processing portion 51 of the control portion 7.

Step S11

[0078] First, in step S11, the control portion 7 executes a second determination process of determining whether or not the elapsed time from the start of the drive of the motor 27A immediately before the reference time at which the drive of the motor 27A is first started during the execution of the conveying processing to the reference time exceeds the length X10 of the determination period Z10. The process of step S11 is an example of the second determination step of the present disclosure, and is executed by the second determination processing portion 54 of the control portion 7.

[0079] Specifically, the control portion 7 executes the second determination process based on the recording result of the change in the drive state of the motor 27A by the recording processing portion 51.

Step S12

[0080] In step S12, the control portion 7 branches the processing in accordance with the result of the second determination process. Specifically, when it is determined in the second determination process that the elapsed time from the start of the drive of the motor 27A immediately before the reference time exceeds the length X10 of the determination period Z10 (Yes in S12), the control portion 7 shifts the processing to step S21. In addition, when it is determined in the second determination process that the elapsed time from the start of the drive of the motor 27A immediately before the reference time does not exceed the length X10 of the determination period Z10 (No in S12), the control portion 7 shifts the processing to step S13.

Step S13

[0081] In step S13, the control portion 7 determines whether or not the conveying processing has been completed.

[0082] Here, when the control portion 7 determines that the conveying processing has been completed (Yes in S13), the control portion 7 terminates the drive control processing. In addition, when the conveying processing has not been completed (No in S13), the control portion 7 shifts the processing to step S14.

Step S14

[0083] In step S14, the control portion 7 determines whether or not the determination timing has arrived.

[0084] Specifically, the control portion 7 determines that the determination timing has arrived when 10 seconds have elapsed since the last arrival of the determination timing (or the reference time).

[0085] Here, when the control portion 7 determines that the determination timing has arrived (Yes in S14), the control portion 7 shifts the processing to step S15. In addition, when the determination timing has not arrived (No in S14), the control portion 7 shifts the processing to step S13.

Step S15

[0086] In step S15, the control portion 7 executes a first determination process of determining whether or not the total drive time of the motor 27A in the determination period Z10 up to the last arrival of the determination timing exceeds the first time X1. The process of step S15 is an example of the first determination step of the present disclosure, and is executed by the first determination processing portion 52 of the control portion 7.

[0087] Specifically, the control portion 7 executes the first determination process based on the recording result of the change in the drive state of the motor 27A by the recording processing portion 51.

Step S16

[0088] In step S16, the control portion 7 branches the processing in accordance with the result of the first determination process. Specifically, when it is determined in the first determination process that the total drive time of the motor 27A in the determination period Z10 up to the last arrival of the determination timing exceeds the first time X1 (Yes in S16), the control portion 7 shifts the processing to step S17. When it is determined in the first determination process that the total drive time of the motor 27A in the determination period Z10 up to the last arrival of the determination timing does not exceed the first time X1 (No in S16), the control portion 7 shifts the processing to step S13.

Step S17

[0089] In step S17, the control portion 7 restricts the drive of the motor 27A. The process of step S17 is an example of the first restriction step of the present disclosure, and is executed by the first restriction processing portion 53 of the control portion 7.

[0090] Specifically, the control portion 7 switches the conveying mode of the document sheet at the ADF 1 from the normal conveying mode to the low-speed conveying mode.

Step S18

[0091] In step S18, the control portion 7 determines whether or not the conveying processing has been completed.

[0092] Here, when the control portion 7 determines that the conveying processing has been completed (Yes in S18), the control portion 7 terminates the drive control processing. In addition, when the conveying processing has not been completed (No in S18), the control portion 7 waits for the conveying processing to be completed in step S18.

[0093] It is noted that the control portion 7 may lift the restriction on the drive of the motor 27A when the total drive time of the motor 27A in the determination period Z10 decreases to the third time or less or when the low-speed conveying mode continues beyond a predetermined time while waiting for the completion of the conveying processing.

Step S21

[0094] In step S21, the control portion 7 determines whether or not the conveying processing has been completed.

[0095] Here, when the control portion 7 determines that the conveying processing has been completed (Yes in S21), the control portion 7 terminates the drive control processing. In addition, when the conveying processing has not been completed (No in S21), the control portion 7 shifts the processing to step S22.

Step S22

[0096] In step S22, the control portion 7 determines whether or not the drive time of the motor 27A from the reference time exceeds the second time X2.

[0097] Here, when it is determined that the drive time of the motor 27A from the reference time exceeds the second time X2 (Yes in S22), the control portion 7 shifts the processing to step S23. When the drive time of the motor 27A from the reference time does not exceed the second time X2 (No in S22), the control portion 7 shifts the processing to step S21.

Step S23

[0098] In step S23, the control portion 7 restricts the drive of the motor 27A as in step S17. Here, the process of step S23 is an example of the second restriction step of the present disclosure, and is executed by the second restriction processing portion 55 of the control portion 7.

Step S24

[0099] In step S24, the control portion 7 determines whether or not the conveying processing has been completed.

[0100] Here, when the control portion 7 determines that the conveying processing has been completed (Yes in S24), the control portion 7 terminates the drive control processing. In addition, when the conveying processing has not been completed (No in S24), the control portion 7 waits for the conveying processing to be completed in step S24.

[0101] It is noted that the control portion 7 may lift the restriction on the drive of the motor 27A when the total drive time of the motor 27A in the determination period Z10 decreases to the third time or less or when the low-speed conveying mode continues beyond a predetermined time while waiting for the completion of the conveying processing.

[0102] As described above, in the image forming apparatus 100, the drive of the motor 27A is controlled based on the total drive time of the motor 27A in the determination period Z10. As a result, it is possible to suppress the temperature rise of the motor 27A with a simpler configuration than the configuration in which the drive of the motor 27A is controlled based on the predicted temperature of the motor 27A.

Appendixes of Invention

[0103] The following are appendixes to the overview of the invention extracted from the above embodiment. It is noted that the structures and processing functions to be described in the following appendixes can be selected and combined arbitrarily.

Appendix 1

[0104] A sheet conveying device comprising: a motor used to convey a sheet; a first determination processing portion configured to determine, each time a predetermined determination timing arrives during drive of the motor, whether or not a total drive time of the motor in a predetermined determination period up to the arrival of the determination timing exceeds a predetermined first time; a first restriction processing portion configured to restrict the drive of the motor when the first determination processing portion determines that the total drive time exceeds the first time; a second determination processing portion configured to determine whether or not an elapsed time from start of the drive of the motor immediately before a reference time at which the drive of the motor is started to the reference time exceeds a length of the determination period; and a second restriction processing portion configured to restrict the drive of the motor when the second determination processing portion determines that the elapsed time exceeds the length of the determination period and a time of the drive of the motor from the reference time exceeds a second time that is shorter than the first time.

Appendix 2

[0105] The sheet conveying device according to Appendix 1, wherein the second time is set based on a difference between the length of the determination period and a stop time of the motor up to the reference time, and the first time.

Appendix 3

[0106] An image processing apparatus comprising: the sheet conveying device according to Appendix 1 or 2; and an image reading portion configured to read an image of the sheet conveyed by the sheet conveying device.

Appendix 4

[0107] A drive control method executed in a sheet conveying device including a motor used to convey a sheet, comprising: a first determination step of determining, each time a predetermined determination timing arrives during drive of the motor, whether or not a total drive time of the motor in a predetermined determination period up to the arrival of the determination timing exceeds a predetermined first time; a first restriction step of restricting the drive of the motor when it is determined in the first determination step that the total drive time exceeds the first time; a second determination step of determining whether or not an elapsed time from start of the drive of the motor immediately before a reference time at which the drive of the motor is started to the reference time exceeds a length of the determination period; and a second restriction step of restricting the drive of the motor when it is determined in the second determination step that the

elapsed time exceeds the length of the determination period and a time of the drive of the motor from the reference time exceeds a second time that is shorter than the first time.

[0108] It is to be understood that the embodiments herein are illustrative and not restrictive, since the scope of the disclosure is defined by the appended claims rather than by the description preceding them, and all changes that fall within metes and bounds of the claims, or equivalence of such metes and bounds thereof are therefore intended to be embraced by the claims.

Claims

1. A sheet conveying device comprising: a motor used to convey a sheet; a first determination processing portion configured to determine, each time a predetermined determination timing arrives during drive of the motor, whether or not a total drive time of the motor in a predetermined determination period up to the arrival of the determination timing exceeds a predetermined first time; a first restriction processing portion configured to restrict the drive of the motor when the first determination processing portion determines that the total drive time exceeds the first time; a second determination processing portion configured to determine whether or not an elapsed time from start of the drive of the motor immediately before a reference time at which the drive of the motor is started to the reference time exceeds a length of the determination period; and a second restriction processing portion configured to restrict the drive of the motor when the second determination processing portion determines that the elapsed time exceeds the length of the determination period and a time of the drive of the motor from the reference time exceeds a second time that is shorter than the first time.
 2. The sheet conveying device according to claim 1, wherein the second time is set based on a difference between the length of the determination period and a stop time of the motor up to the reference time, and the first time.
 3. An image processing apparatus comprising: the sheet conveying device according to claim 1; and an image reading portion configured to read an image of the sheet conveyed by the sheet conveying device.
 4. A drive control method executed in a sheet conveying device including a motor used to convey a sheet, comprising: a first determination step of determining, each time a predetermined determination timing arrives during drive of the motor, whether or not a total drive time of the motor in a predetermined determination period up to the arrival of the determination timing exceeds a predetermined first time; a first restriction step of restricting the drive of the motor when it is determined in the first determination step that the total drive time exceeds the first time; a second determination step of determining whether or not an elapsed time from start of the drive of the motor immediately before a reference time at which the drive of the motor is started to the reference time exceeds a length of the determination period; and a second restriction step of restricting the drive of the motor when it is determined in the second determination step that the elapsed time exceeds the length of the determination period and a time of the drive of the motor from the reference time exceeds a second time that is shorter than the first time.
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